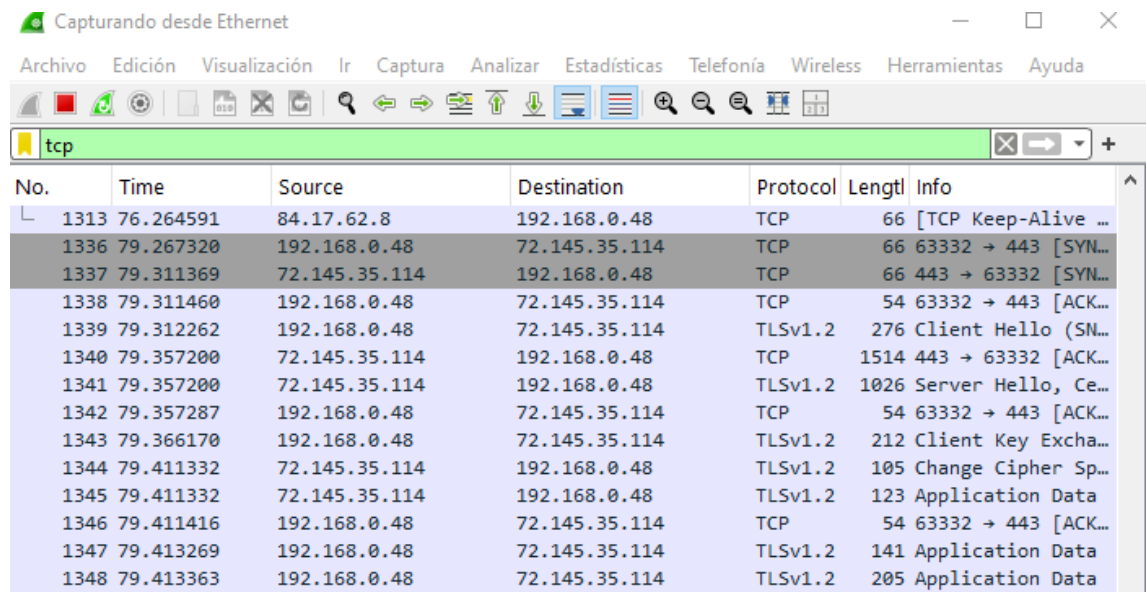


EJERCICIOS DE WIRESHARK

Protocolos

1. Todo el tráfico TCP:

Tcp

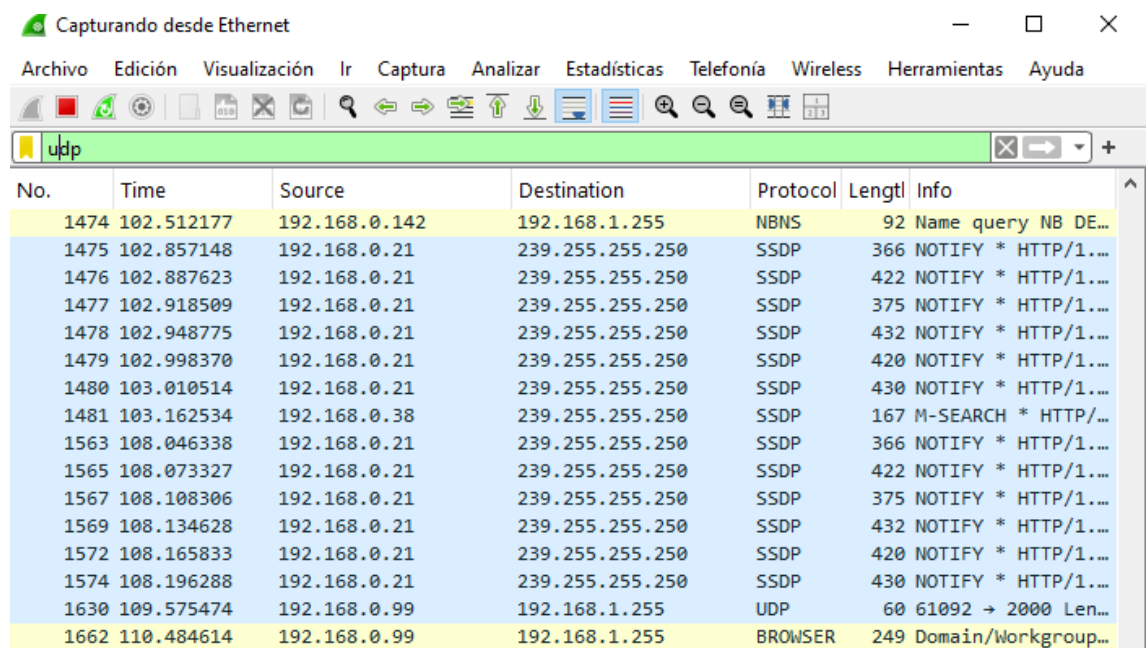


The screenshot shows the Wireshark interface with the 'tcp' filter applied. The packet list displays various TCP and TLSv1.2 packets. The packet details pane is empty, and the packet bytes pane shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
1313	76.264591	84.17.62.8	192.168.0.48	TCP	66	[TCP Keep-Alive ...
1336	79.267320	192.168.0.48	72.145.35.114	TCP	66	63332 → 443 [SYN...
1337	79.311369	72.145.35.114	192.168.0.48	TCP	66	443 → 63332 [SYN...
1338	79.311460	192.168.0.48	72.145.35.114	TCP	54	63332 → 443 [ACK...
1339	79.312262	192.168.0.48	72.145.35.114	TLSv1.2	276	Client Hello (SN...
1340	79.357200	72.145.35.114	192.168.0.48	TCP	1514	443 → 63332 [ACK...
1341	79.357200	72.145.35.114	192.168.0.48	TLSv1.2	1026	Server Hello, Ce...
1342	79.357287	192.168.0.48	72.145.35.114	TCP	54	63332 → 443 [ACK...
1343	79.366170	192.168.0.48	72.145.35.114	TLSv1.2	212	Client Key Excha...
1344	79.411332	72.145.35.114	192.168.0.48	TLSv1.2	105	Change Cipher Sp...
1345	79.411332	72.145.35.114	192.168.0.48	TLSv1.2	123	Application Data
1346	79.411416	192.168.0.48	72.145.35.114	TCP	54	63332 → 443 [ACK...
1347	79.413269	192.168.0.48	72.145.35.114	TLSv1.2	141	Application Data
1348	79.413363	192.168.0.48	72.145.35.114	TLSv1.2	205	Application Data

2. Todo el tráfico UDP:

Udp

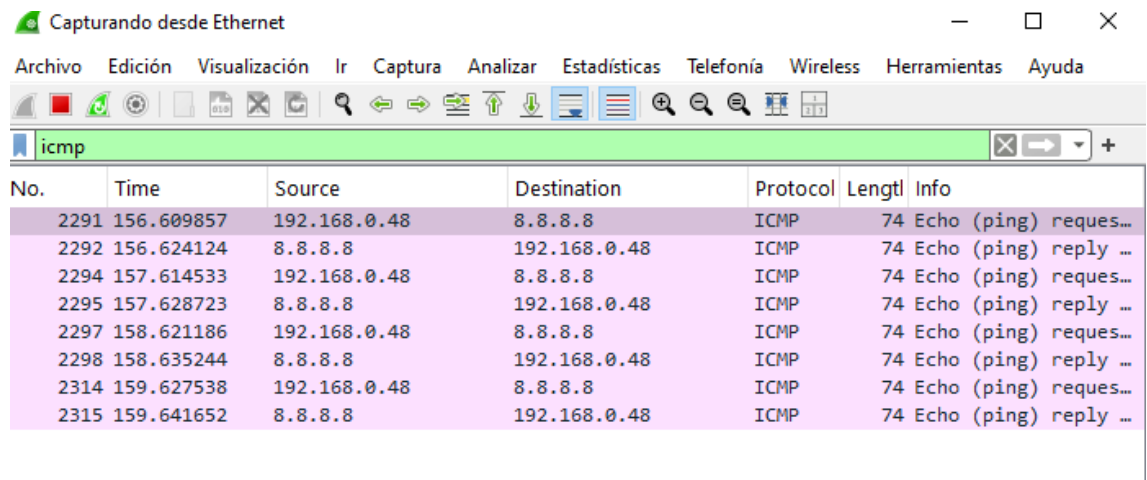


The screenshot shows the Wireshark interface with the 'udp' filter applied. The packet list displays various UDP and NBNS packets. The packet details pane is empty, and the packet bytes pane shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
1474	102.512177	192.168.0.142	192.168.1.255	NBNS	92	Name query NB DE...
1475	102.857148	192.168.0.21	239.255.255.250	SSDP	366	NOTIFY * HTTP/1...
1476	102.887623	192.168.0.21	239.255.255.250	SSDP	422	NOTIFY * HTTP/1...
1477	102.918509	192.168.0.21	239.255.255.250	SSDP	375	NOTIFY * HTTP/1...
1478	102.948775	192.168.0.21	239.255.255.250	SSDP	432	NOTIFY * HTTP/1...
1479	102.998370	192.168.0.21	239.255.255.250	SSDP	420	NOTIFY * HTTP/1...
1480	103.010514	192.168.0.21	239.255.255.250	SSDP	430	NOTIFY * HTTP/1...
1481	103.162534	192.168.0.38	239.255.255.250	SSDP	167	M-SEARCH * HTTP/...
1563	108.046338	192.168.0.21	239.255.255.250	SSDP	366	NOTIFY * HTTP/1...
1565	108.073327	192.168.0.21	239.255.255.250	SSDP	422	NOTIFY * HTTP/1...
1567	108.108306	192.168.0.21	239.255.255.250	SSDP	375	NOTIFY * HTTP/1...
1569	108.134628	192.168.0.21	239.255.255.250	SSDP	432	NOTIFY * HTTP/1...
1572	108.165833	192.168.0.21	239.255.255.250	SSDP	420	NOTIFY * HTTP/1...
1574	108.196288	192.168.0.21	239.255.255.250	SSDP	430	NOTIFY * HTTP/1...
1630	109.575474	192.168.0.99	192.168.1.255	UDP	60	61092 → 2000 Len...
1662	110.484614	192.168.0.99	192.168.1.255	BROWSER	249	Domain/Workgroup...

3. Tráfico ICMP (ping):

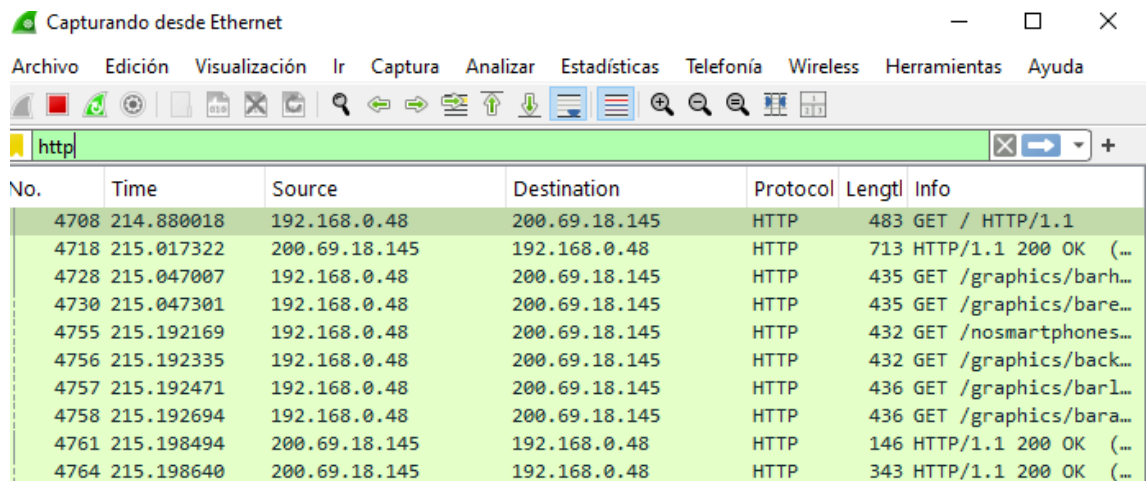
Icmp



No.	Time	Source	Destination	Protocol	Length	Info
2291	156.609857	192.168.0.48	8.8.8.8	ICMP	74	Echo (ping) request
2292	156.624124	8.8.8.8	192.168.0.48	ICMP	74	Echo (ping) reply
2294	157.614533	192.168.0.48	8.8.8.8	ICMP	74	Echo (ping) request
2295	157.628723	8.8.8.8	192.168.0.48	ICMP	74	Echo (ping) reply
2297	158.621186	192.168.0.48	8.8.8.8	ICMP	74	Echo (ping) request
2298	158.635244	8.8.8.8	192.168.0.48	ICMP	74	Echo (ping) reply
2314	159.627538	192.168.0.48	8.8.8.8	ICMP	74	Echo (ping) request
2315	159.641652	8.8.8.8	192.168.0.48	ICMP	74	Echo (ping) reply

4. Tráfico HTTP (puerto 80):

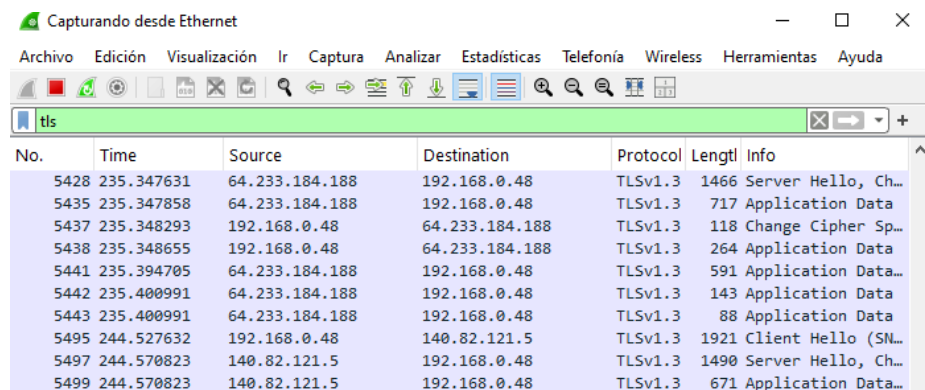
http



No.	Time	Source	Destination	Protocol	Length	Info
4708	214.880018	192.168.0.48	200.69.18.145	HTTP	483	GET / HTTP/1.1
4718	215.017322	200.69.18.145	192.168.0.48	HTTP	713	HTTP/1.1 200 OK (...)
4728	215.047007	192.168.0.48	200.69.18.145	HTTP	435	GET /graphics/barh...
4730	215.047301	192.168.0.48	200.69.18.145	HTTP	435	GET /graphics/bare...
4755	215.192169	192.168.0.48	200.69.18.145	HTTP	432	GET /nosmartphones...
4756	215.192335	192.168.0.48	200.69.18.145	HTTP	432	GET /graphics/back...
4757	215.192471	192.168.0.48	200.69.18.145	HTTP	436	GET /graphics/barl...
4758	215.192694	192.168.0.48	200.69.18.145	HTTP	436	GET /graphics/bara...
4761	215.198494	200.69.18.145	192.168.0.48	HTTP	146	HTTP/1.1 200 OK (...)
4764	215.198640	200.69.18.145	192.168.0.48	HTTP	343	HTTP/1.1 200 OK (...)

5. Tráfico HTTPS (TLS/SSL):

Tls



No.	Time	Source	Destination	Protocol	Length	Info
5428	235.347631	64.233.184.188	192.168.0.48	TLSv1.3	1466	Server Hello, Ch...
5435	235.347858	64.233.184.188	192.168.0.48	TLSv1.3	717	Application Data
5437	235.348293	192.168.0.48	64.233.184.188	TLSv1.3	118	Change Cipher Sp...
5438	235.348655	192.168.0.48	64.233.184.188	TLSv1.3	264	Application Data
5441	235.394705	64.233.184.188	192.168.0.48	TLSv1.3	591	Application Data...
5442	235.400991	64.233.184.188	192.168.0.48	TLSv1.3	143	Application Data
5443	235.400991	64.233.184.188	192.168.0.48	TLSv1.3	88	Application Data
5495	244.527632	192.168.0.48	140.82.121.5	TLSv1.3	1921	Client Hello (SN...
5497	244.570823	140.82.121.5	192.168.0.48	TLSv1.3	1490	Server Hello, Ch...
5499	244.570823	140.82.121.5	192.168.0.48	TLSv1.3	671	Application Data...

6. Tráfico DNS (puerto 53):

Dns

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dns

No.	Time	Source	Destination	Protocol	Length	Info
3854	213.026571	192.168.0.48	80.58.61.254	DNS	88	Standard query 0...
3910	213.040958	80.58.61.254	192.168.0.48	DNS	138	Standard query r...
3944	213.047913	80.58.61.254	192.168.0.48	DNS	104	Standard query r...
4639	214.340982	192.168.0.48	80.58.61.254	DNS	74	Standard query 0...
4640	214.341179	192.168.0.48	80.58.61.254	DNS	74	Standard query 0...
4644	214.349399	192.168.0.48	80.58.61.254	DNS	74	Standard query 0...
4645	214.349628	192.168.0.48	80.58.61.254	DNS	74	Standard query 0...
4649	214.378846	192.168.0.48	80.58.61.250	DNS	74	Standard query 0...
4650	214.379006	192.168.0.48	80.58.61.250	DNS	74	Standard query 0...
4652	214.387510	192.168.0.48	80.58.61.250	DNS	74	Standard query 0...

7. Tráfico ARP:

Arp

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arp

No.	Time	Source	Destination	Protocol	Length	Info
5353	229.817413	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....
5354	229.844741	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....
5355	229.870076	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....
5356	229.898441	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....
5357	229.908126	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....
5358	229.944571	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....
5359	229.980262	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....
5360	229.982259	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....
5361	230.018769	AskeyCompute_4e:9f:...	Broadcast	ARP	60	Who has 192.168....

8. Tráfico DHCP:

Bootp

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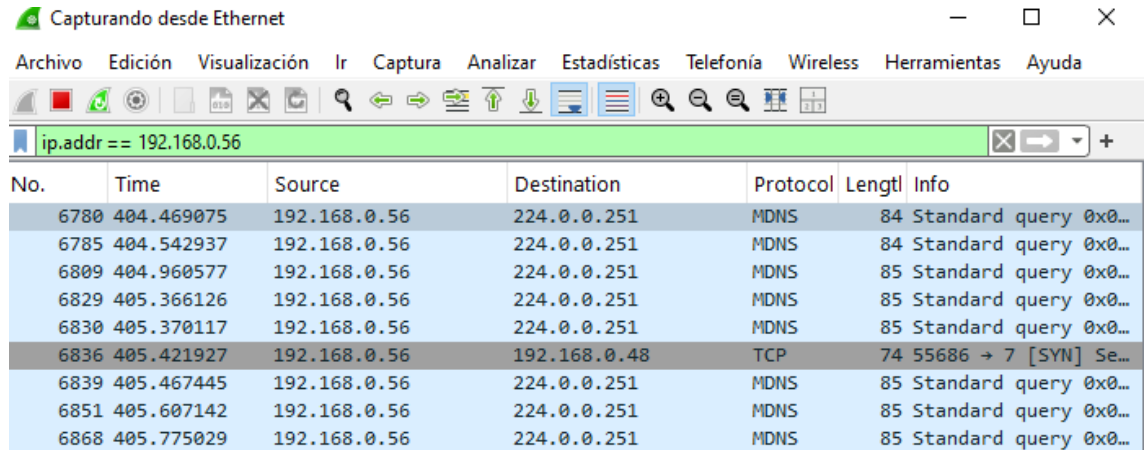
bootp

No.	Time	Source	Destination	Protocol	Length	Info
5788	305.657234	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Tr
5789	305.657740	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Tr

◆ IPs

9. Cualquier tráfico de una IP concreta:

ip.addr == 192.168.0.56



Capturando desde Ethernet

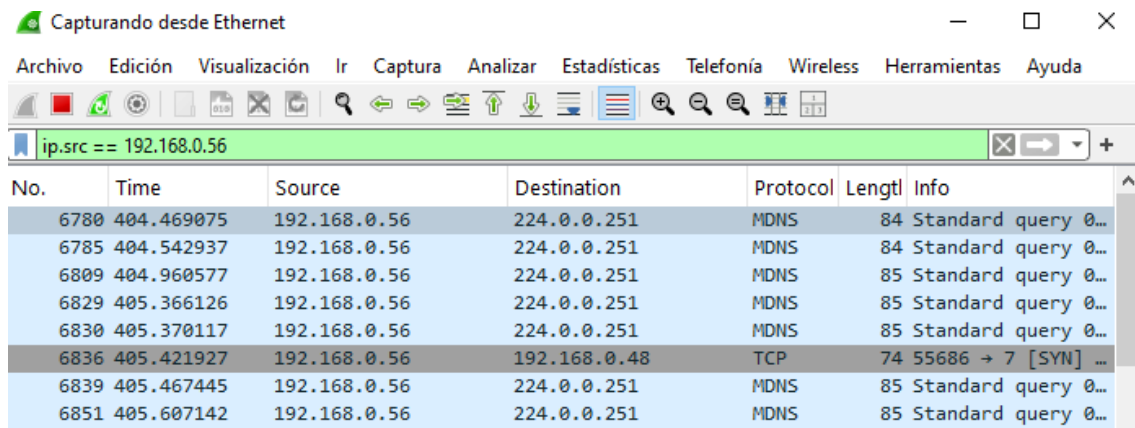
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ip.addr == 192.168.0.56

No.	Time	Source	Destination	Protocol	Length	Info
6780	404.469075	192.168.0.56	224.0.0.251	MDNS	84	Standard query 0x0...
6785	404.542937	192.168.0.56	224.0.0.251	MDNS	84	Standard query 0x0...
6809	404.960577	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0x0...
6829	405.366126	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0x0...
6830	405.370117	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0x0...
6836	405.421927	192.168.0.56	192.168.0.48	TCP	74	55686 → 7 [SYN] Se...
6839	405.467445	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0x0...
6851	405.607142	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0x0...
6868	405.775029	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0x0...

10. Solo tráfico de origen desde una IP:

ip.src == 192.168.0.56



Capturando desde Ethernet

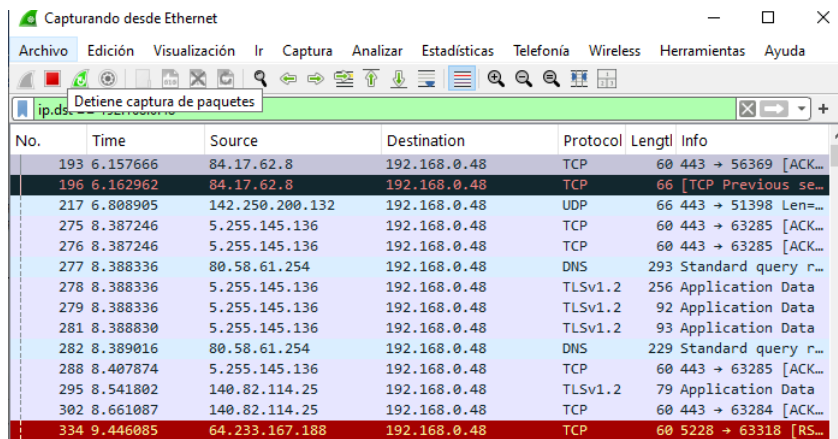
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ip.src == 192.168.0.56

No.	Time	Source	Destination	Protocol	Length	Info
6780	404.469075	192.168.0.56	224.0.0.251	MDNS	84	Standard query 0...
6785	404.542937	192.168.0.56	224.0.0.251	MDNS	84	Standard query 0...
6809	404.960577	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0...
6829	405.366126	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0...
6830	405.370117	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0...
6836	405.421927	192.168.0.56	192.168.0.48	TCP	74	55686 → 7 [SYN] ...
6839	405.467445	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0...
6851	405.607142	192.168.0.56	224.0.0.251	MDNS	85	Standard query 0...

11. Solo tráfico con destino a una IP:

ip.dst == 192.168.0.48



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ip.dst == 192.168.0.48

No.	Time	Source	Destination	Protocol	Length	Info
193	6.157666	84.17.62.8	192.168.0.48	TCP	60	443 → 56369 [ACK...
196	6.162962	84.17.62.8	192.168.0.48	TCP	60	[TCP Previous se...
217	6.808905	142.250.200.132	192.168.0.48	UDP	66	443 → 51398 Len=...
275	8.387246	5.255.145.136	192.168.0.48	TCP	60	443 → 63285 [ACK...
276	8.387246	5.255.145.136	192.168.0.48	TCP	60	443 → 63285 [ACK...
277	8.388336	80.58.61.254	192.168.0.48	DNS	293	Standard query r...
278	8.388336	5.255.145.136	192.168.0.48	TLSv1.2	256	Application Data
279	8.388336	5.255.145.136	192.168.0.48	TLSv1.2	92	Application Data
281	8.388830	5.255.145.136	192.168.0.48	TLSv1.2	93	Application Data
282	8.389016	80.58.61.254	192.168.0.48	DNS	229	Standard query r...
288	8.407874	5.255.145.136	192.168.0.48	TCP	60	443 → 63285 [ACK...
295	8.541802	140.82.114.25	192.168.0.48	TLSv1.2	79	Application Data
302	8.661087	140.82.114.25	192.168.0.48	TCP	60	443 → 63284 [ACK...
334	9.446085	64.233.167.188	192.168.0.48	TCP	60	5228 → 63318 [RS...

12. Tráfico entre dos IP específicas:

ip.src == 192.168.0.48 && ip.dst == 8.8.8.8

Capturando desde Ethernet

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ip.src == 192.168.0.48 && ip.dst == 8.8.8.8

No.	Time	Source	Destination	Protocol	Length	Info
2291	156.609857	192.168.0.48	8.8.8.8	ICMP	74	Echo (ping) request...
2294	157.614533	192.168.0.48	8.8.8.8	ICMP	74	Echo (ping) request...
2297	158.621186	192.168.0.48	8.8.8.8	ICMP	74	Echo (ping) request...
2314	159.627538	192.168.0.48	8.8.8.8	ICMP	74	Echo (ping) request...

◆ Puertos

13. Tráfico en un puerto concreto (ej. 22 – SSH):

tcp.port == 22

Capturando desde Ethernet

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tcp.port == 22

No.	Time	Source	Destination	Protocol	Length	Info
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14. Tráfico en un rango de puertos (ej. 20 a 25):

tcp.port >= 20 && tcp.port <= 25

Capturando desde Ethernet

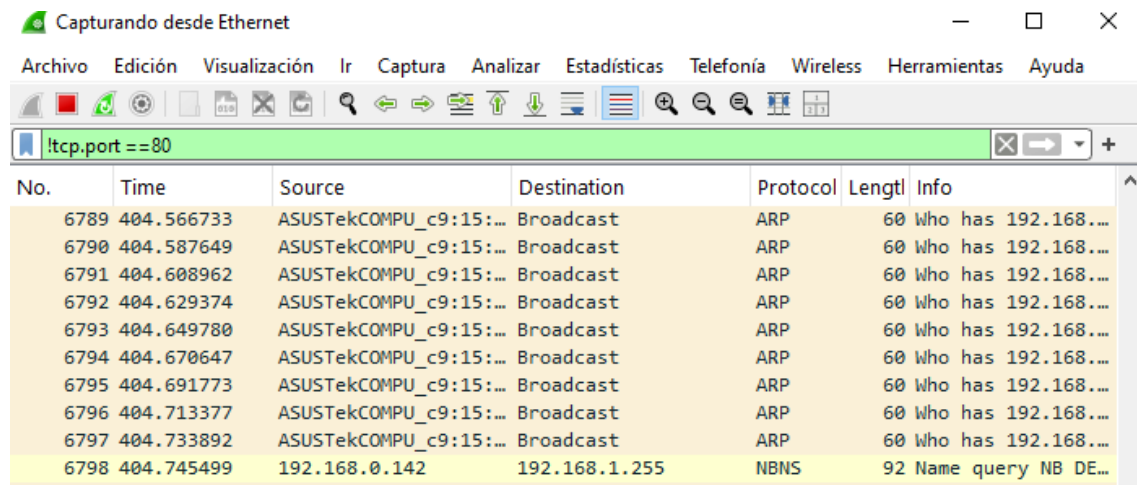
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tcp.port >= 20 && tcp.port <= 25

No.	Time	Source	Destination	Protocol	Length	Info
6836	405.421927	192.168.0.56	192.168.0.48	TCP	74	55686
6929	406.452092	192.168.0.56	192.168.0.48	TCP	74	[TCP R
7008	407.424398	192.168.0.56	192.168.0.48	TCP	74	55700
7075	408.432393	192.168.0.56	192.168.0.48	TCP	74	[TCP R

15. Tráfico NO en un puerto concreto (ej. todo excepto 80):

!tcp.port == 80



Capturando desde Ethernet

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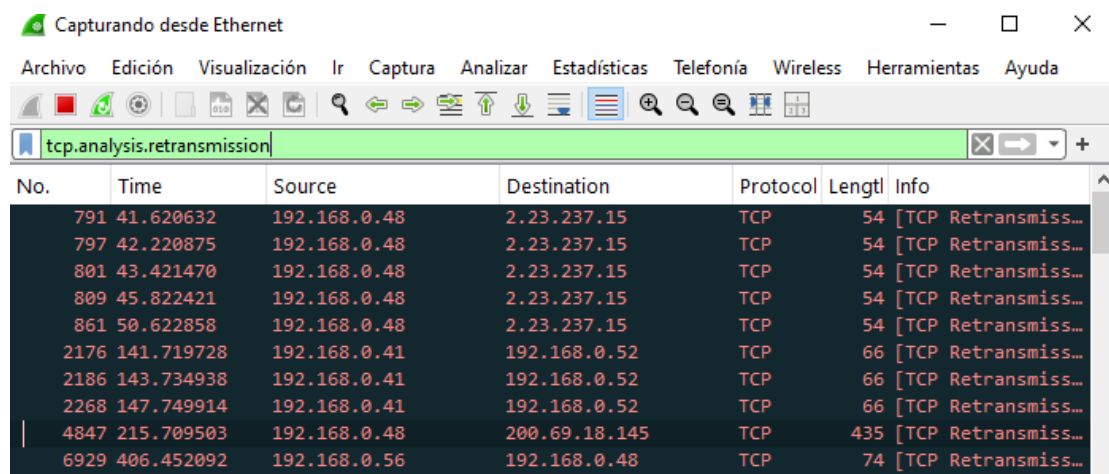
!tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
6789	404.566733	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6790	404.587649	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6791	404.608962	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6792	404.629374	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6793	404.649780	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6794	404.670647	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6795	404.691773	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6796	404.713377	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6797	404.733892	ASUSTekCOMPU_c9:15:...	Broadcast	ARP	60	Who has 192.168...
6798	404.745499	192.168.0.142	192.168.1.255	NBNS	92	Name query NB DE...

◆ TCP avanzado

16. Retransmisiones TCP:

tcp.analysis.retransmission



Capturando desde Ethernet

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tcp.analysis.retransmission

No.	Time	Source	Destination	Protocol	Length	Info
791	41.620632	192.168.0.48	2.23.237.15	TCP	54	[TCP Retransmiss...
797	42.220875	192.168.0.48	2.23.237.15	TCP	54	[TCP Retransmiss...
801	43.421470	192.168.0.48	2.23.237.15	TCP	54	[TCP Retransmiss...
809	45.822421	192.168.0.48	2.23.237.15	TCP	54	[TCP Retransmiss...
861	50.622858	192.168.0.48	2.23.237.15	TCP	54	[TCP Retransmiss...
2176	141.719728	192.168.0.41	192.168.0.52	TCP	66	[TCP Retransmiss...
2186	143.734938	192.168.0.41	192.168.0.52	TCP	66	[TCP Retransmiss...
2268	147.749914	192.168.0.41	192.168.0.52	TCP	66	[TCP Retransmiss...
4847	215.709503	192.168.0.48	200.69.18.145	TCP	435	[TCP Retransmiss...
6929	406.452092	192.168.0.56	192.168.0.48	TCP	74	[TCP Retransmiss...

17. Paquetes TCP con errores:

tcp.analysis.flags

Capturando desde Ethernet

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

tcp.analysis.flags

No.	Time	Source	Destination	Protocol	Length	Info
2685	186.410498	192.168.0.48	84.17.62.8	TCP	55	[TCP Keep-Alive]...
2687	186.425051	84.17.62.8	192.168.0.48	TCP	66	[TCP Keep-Alive ...
2729	190.045189	192.168.0.48	142.251.5.188	TCP	55	[TCP Keep-Alive]...
3169	196.425599	192.168.0.48	84.17.62.8	TCP	55	[TCP Keep-Alive]...
3170	196.439673	84.17.62.8	192.168.0.48	TCP	66	[TCP Keep-Alive ...
3256	204.013139	192.168.0.48	185.199.110.133	TCP	55	[TCP Keep-Alive]...
3257	204.027653	185.199.110.133	192.168.0.48	TCP	66	[TCP Keep-Alive ...
3262	204.377258	192.168.0.48	140.82.112.25	TCP	55	[TCP Keep-Alive]...
3263	204.493503	140.82.112.25	192.168.0.48	TCP	66	[TCP Keep-Alive ...
3440	206.440539	192.168.0.48	84.17.62.8	TCP	55	[TCP Keep-Alive]...
3442	206.454576	84.17.62.8	192.168.0.48	TCP	66	[TCP Keep-Alive ...
4847	215.709503	192.168.0.48	200.69.18.145	TCP	435	[TCP Retransmiss...

18. Solo handshakes TCP (SYN):

tcp.flags.syn == 1 && tcp.flags.ack == 0

Capturando desde Ethernet

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

tcp.flags.syn == 1 && tcp.flags.ack == 0

No.	Time	Source	Destination	Protocol	Length	Info
2878	191.203516	192.168.0.48	185.199.111.154	TCP	66	63342 → 443 [SYN...
3094	192.076670	192.168.0.48	140.82.121.6	TCP	66	63343 → 443 [SYN...
3301	205.467237	192.168.0.48	142.250.184.14	TCP	66	63344 → 443 [SYN...
3380	205.908040	192.168.0.48	142.250.184.161	TCP	66	63345 → 443 [SYN...
3734	212.955371	192.168.0.48	142.250.185.8	TCP	66	63346 → 443 [SYN...
4658	214.438703	192.168.0.48	200.69.18.145	TCP	66	63347 → 80 [SYN]...
4659	214.438930	192.168.0.48	200.69.18.145	TCP	66	63348 → 80 [SYN]...
4660	214.439288	192.168.0.48	200.69.18.145	TCP	66	63349 → 443 [SYN...
4727	215.046914	192.168.0.48	200.69.18.145	TCP	66	63350 → 80 [SYN]...
4729	215.047234	192.168.0.48	200.69.18.145	TCP	66	63351 → 80 [SYN]...
4731	215.047472	192.168.0.48	200.69.18.145	TCP	66	63352 → 80 [SYN]...
4732	215.047652	192.168.0.48	200.69.18.145	TCP	66	63353 → 80 [SYN]...
5044	220.345771	192.168.0.47	192.168.0.43	TCP	66	60029 → 7680 [SY...

19. Conexiones TCP reseteadas (RST):

tcp.flags.reset == 1

Capturando desde Ethernet

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tcp.flags.reset == 1

No.	Time	Source	Destination	Protocol	Length	Info
1436	99.759425	142.251.5.188	192.168.0.48	TCP	60	5228 → 63321 [RS...
1462	100.025969	192.168.0.48	140.82.113.22	TCP	54	63276 → 443 [RST...
1468	101.181451	192.168.0.48	140.82.121.5	TCP	54	63327 → 443 [RST...
2242	144.923134	142.251.5.188	192.168.0.48	TCP	60	5228 → 63333 [RS...
2397	179.368024	150.171.69.254	192.168.0.48	TCP	60	443 → 63322 [RST...
2402	180.879612	13.107.3.254	192.168.0.48	TCP	60	443 → 63323 [RST...
2410	182.309547	20.111.38.59	192.168.0.48	TCP	60	443 → 63324 [RST...
2460	184.388713	192.168.0.48	185.199.111.154	TCP	54	63279 → 443 [RST...
2730	190.081746	142.251.5.188	192.168.0.48	TCP	60	5228 → 63337 [RS...
3165	195.524533	172.202.65.254	192.168.0.48	TCP	60	443 → 63331 [RST...
3179	197.287617	13.107.6.254	192.168.0.48	TCP	60	443 → 63329 [RST...
3223	201.190268	52.123.129.254	192.168.0.48	TCP	60	443 → 63330 [RST...
5098	221.846279	192.168.0.48	140.82.121.3	TCP	54	63340 → 443 [RST...

◆ Otros filtros útiles

20. Mostrar solo paquetes que contienen un texto (ej. “login”):

frame contains "login"

